MIL-A-60889(MU)
9 April 1968

MILITARY SPECIFICATION
ACTUATOR, EXPLOSIVE, ROTARY 1000-45
REMANUFACTURED METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Actuator, Explosive, Rotary 1000-45, Remanufactured Metal Parts Assembly. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-413	- Visual Inspection Guide for Rubber O-Rings
MIL-STD-639	- Surface Roughness Values Preferred for Army Ordnance Materiel
MIL-STD-1167	- Ammunition Data Card

DRAWINGS

Munitions Command

C9218443	- Carton, Packing Ammunition, for Rotary Actuator, 1000 or 1925 series
C9218444	- Box, Packing Ammunition, for Rotary Actuator, 1000 or 1925 series

FSC 1377

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2.1 DRAWINGS (Cont'd)

Talley Industries (Code Ident. No. 98708)
1000-45 - Rotary Actuator Assembly

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Hydrostatic pressure.- The body, Part Number 1072 (Talley) shall withstand a hydrostatic pressure of 7500 pounds per square inch (psi) for 15 seconds, minimum, without deformation or mechanical failure. (See 6.1).

3.3 Metal parts assembly.- Parts shall be combined to produce a metal parts assembly which conforms to the requirements of the assembly. All threaded joints shall be assembled hand-tight.

3.3.1 The remanufactured actuator metal parts assembly shall contain new parts in accordance with Table I.

TABLE I

Name of Part	Part Number
End Closure Assembly	1000-132 (Talley)
Pin, Shear	1019 (Talley)
Packing, O-Rings	MS9021-012
Packing, O-Rings	MS9021-018
Packing, O-Rings	MS9021-020
Packing, O-Rings	MS9021-032
Packing, O-Rings	MS9021-009
Packing, O-Rings	MS9021-014
Stud, Shear	1003 (Talley)
Strap, Assembly	1000-104 (Talley)
Pellet	1006 (Talley)

3.4 Protective finish.- Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component, regardless of the configuration. Remanufactured exterior components shall have the existing protective finish stripped and the specified protective finish reapplied (See 6.2 and 6.3).

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3.5 Residual magnetism.- The metal parts assembly shall not deflect the indicator of the approved compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches (See 6.4).

3.6 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.6.1 Metal defects.- The part shall be free from cracks, splits, or other metal defects.

3.6.2 Burr.- The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.6.3 Foreign matter.- The part or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the Quality Assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

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4.2.1.2 Size of lot.- For the purpose of identification and shipment, the shipping lots of metal parts assemblies shall consist of 1200 Actuators, maximum, exclusive of test samples, provided the homogeneity of remanufacturing process has been retained throughout the inspection lot. A smaller lot may be shipped upon approval of the contracting officer.

4.2.2 Examination.- The Ordnance Standard Finish Blocks prescribed in MIL-STD-639 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

4.2.2.1 Reclaimed parts.- Examination of reusable parts, which require no reprocessing shall be limited to inspection of defects that require visual type inspection. One hundred percent examination shall be performed for all such defects. All nonconforming parts shall be rejected.

4.2.2.2 Reprocessed parts.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables II and III. For an individual defect (single characteristic) the criteria in Table II shall be used. For total defects within a major or minor group, the criteria in Table III shall be used. All nonconforming material shall be rejected.

TABLE II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

TABLE III

Total Allowable Defects Per Sample Size

Sample Sizes for Major Defects	110	150	225	300
Number of Defects in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over 43	6	8	11	14

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TABLE III (Cont'd)

Sample Sizes for Minor Defects	110	150	225	300
Number of Defects in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over 48	8	11	17	20

4.2.2.3 Breech, Drawing 1001, covering a detail of Drawing 1000-45

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of thread, max.	Gage
102. Minor diameter of thread, max.	Gage
103. Shear pin hole	Gage
104. Metal defective	Visual
105. Protective finish inadequate or defective	Visual
106. Surface finish improper	Visual
107. Burr	Visual
108. Foreign matter	Visual
MINOR	
None defined	

4.2.2.4 Breech, Subassembly, Drawing 1022 covering a detail of Drawing 1000-45

CRITICAL	
1. Any part missing	Gage

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4.2.2.4 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Improper assembly	Visual
102. "O" ring not lubricated	Visual
103. Threads damaged	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
201. Pellet missing, loose or improperly inserted	Visual and Manual

4.2.2.5 Ring, Firing Port, Drawing 1002, covering a detail of Drawing 1000-45

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of thread, max.	Gage
102. Major diameter of thread, min.	Gage
103. Metal defective	Visual
104. Protective finish inadequate or defective	Visual
105. Surface finish on bore and face of port improper	Visual
106. Burr	Visual
107. Foreign matter	Visual
MINOR	
None defined	

4.2.2.6 Support, Bearing, Drawing 1004 covering a detail of Drawing 1000-45

CRITICAL	
None defined	

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4.2.2.6 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Pitch diameter of thread, max.	Gage
102. Major diameter of thread, min.	Gage
103. Pitch diameter of thread, max., 4 holes	Gage
104. Major diameter of thread, min., 4 holes	Gage
105. Pitch diameter of thread, max. of small hole	Gage
106. Major diameter of thread, min. of small hole	Gage
107. Threads damaged	Visual
108. Metal defective	Visual
109. Protective finish inadequate or defective	Visual
110. Burr	Visual
111. Foreign matter	Visual
MINOR	
None defined	

4.2.2.7 Bracket, Mounting, Drawing 1082 covering a detail of Drawing 1000-45

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of thread, max.	Gage
102. Major diameter of thread, min.	Gage
103. Threads damaged	Visual
104. Metal defective	Visual
105. Protective finish inadequate or defective	Visual
106. Burr	Visual
107. Foreign matter	Visual
MINOR	
None defined	

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4.2.2.8 Pin, Firing, Drawing 1007 covering a detail of Drawing 1000-45

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Metal defective	Visual
102. Protective finish inadequate	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

4.2.2.9 Plug, Blowout, Drawing 1015 covering a detail of Drawing 1000-45

CRITICAL	
None defined	
MAJOR	
101. Metal defective	Visual
102. Protective finish inadequate or defective	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

4.2.2.10 Washer, Drawing 1017 covering a detail of Drawing 1000-45

CRITICAL	
None defined	

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4.2.2.10 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Material defective	Visual
102. Protective finish inadequate or defective	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

4.2.2.11 Retainer, Cartridge, Drawing 1018 covering a detail of Drawing 1000-45

CRITICAL	
None defined	
MAJOR	
101. Metal	Visual
102. Protective finish inadequate or defective	Visual
103. Surface finish improper	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
None defined	

4.2.2.12 Head, Piston, Drawing 1021 covering a detail of Drawing 1000-45

CRITICAL	
None defined	
MAJOR	
101. Threads damaged	Visual
102. Protective finish inadequate or defective	Visual
103. Surface finish improper	Visual
104. Burr	Visual
105. Foreign matter	Visual

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4.2.2.12 (Cont'd)

Defect	Method of Inspection

MINOR

None defined

4.2.2.13 Ball, Screw and Piston Assembly, Drawing 1024 covering a detail of Drawing 1000-45

CRITICAL

None defined

MAJOR

101.	Metal defective	Visual
102.	Any part missing	Visual
103.	Improper assembly	Visual
104.	Mounting hole threads damaged	Visual
105.	Strap clamp screw threads damaged	Visual
106.	Lock pin loose, missing	Manual and Visual
107.	"O" ring not lubricated	Visual
108.	Head piston loose	Manual and Visual
109.	Burr	Visual
110.	Foreign matter	Visual

MINOR

None defined

4.2.2.14 Clamp, Strap, Drawing 1032 covering a detail of Drawing 1000-45

CRITICAL

None defined

MAJOR

101.	Metal defective	Visual
102.	Protective finish inadequate or defective	Visual
103.	Burr	Visual
104.	Foreign matter	Visual

MINOR

None defined

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4.2.2.15 Body, Drawing 1072 covering a detail of Drawing 1000-45

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Pitch diameter of thread, max.	Gage
102. Major diameter of thread, min.	Gage
103. Threads damaged	Visual
104. Surface finish improper	Visual
105. Protective finish inadequate or defective	Visual
106. Burr	Visual
107. Foreign matter	Visual
MINOR	
None defined	

4.2.2.16 Metal Parts Assembly, Drawing 1000-45

CRITICAL	
None defined	
MAJOR	
101. Any part missing except as classified	Visual
102. Improper assembly	Visual
103. "O" rings not lubricated	Visual
104. Threads damaged	Visual
105. Burr	Visual
106. Foreign matter	Visual
MINOR	
201. Shipping plug missing	Visual
202. Protective finish inadequate or defective	Visual

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4.2.2.17 Packing in Fiber Carton, Drawing C9218443

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Sealing strips missing, loose or improperly applied	Visual
202. Contents loose	Manual
203. Flaps loose or distorted	Visual
204. Cut or damaged through all impregnated layers	Visual
205. Marking missing, incorrect or illegible	Visual

4.2.2.18 Packing in Wooden Box, Drawing C9218444

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Manual and Visual
203. Strapping missing, loose, broken or improperly assembled	Manual and Visual
204. Contents of box exposed	Visual
205. Marking missing, incorrect or unidentifiable	Visual
206. Number of units incorrect	Visual

4.2.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.3.

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4.2.3.1 Hydrostatic pressure.- Each actuator body assembly shall be subjected to the hydrostatic pressure test. Failure of any item to comply with the requirements of 3.2 shall be cause for rejection of the subassembly.

4.2.3.2 Protective finish, replated components.- The number of specimens and acceptance criteria for protective finish shall be in accordance with the applicable Military Standard listed on the drawing.

4.2.3.3 Residual magnetism.- Each actuator shall be tested for excessive residual magnetism. Failure of any actuator to comply with the requirement of 3.5 shall be cause for rejection of the actuator.

4.2.4 Inspection equipment.- The examination and tests shall be made using equipment and gages approved by the Government.

4.3 Test methods and procedures.

4.3.1 Hydrostatic pressure.- Prior to the application of protective finish each subassembly shall be mounted in a device with all openings properly sealed and subjected to the hydrostatic pressure to determine compliance with the requirements of 3.2.

4.3.2 Protective finish, replated components.- Test procedures for protective finish shall be in accordance with the applicable Military Standards referenced on the drawing.

4.3.3 Residual magnetism.- The assembly shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the assembly six (6) inches from and at the same height as the compass, in a north-south horizontal position (breech end at south) with the compass set in an east-west heading as shown in Figure 1. The assembly shall be moved slowly past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the assembly.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking - Level A.- Packing and marking shall be in accordance with drawings C9218443 and C9218444.

5.2 Data cards.- Data card information shall be as specified in MIL-STD-1167.

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6. NOTES

6.1 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.2 Remanufactured exterior components are listed in Table IV.

TABLE IV

Name of Part	Part Number (Talley)
Bracket, Mounting	1082
Breech Subassembly	1022
Ring, Firing Port	1002
Support, Bearing	1004
Body	1072

6.3 Nylon pellets.- Components which contain nylon pellets and are to be replated shall have the pellets removed and new pellets installed, after plating, in accordance with the applicable drawing.

6.4 When remanufacturing and loading of assembly is performed in an integrated plant, the metal parts assembly requirement may be waived and the residual magnetism requirement performed on the loaded assembly only.

Custodian:

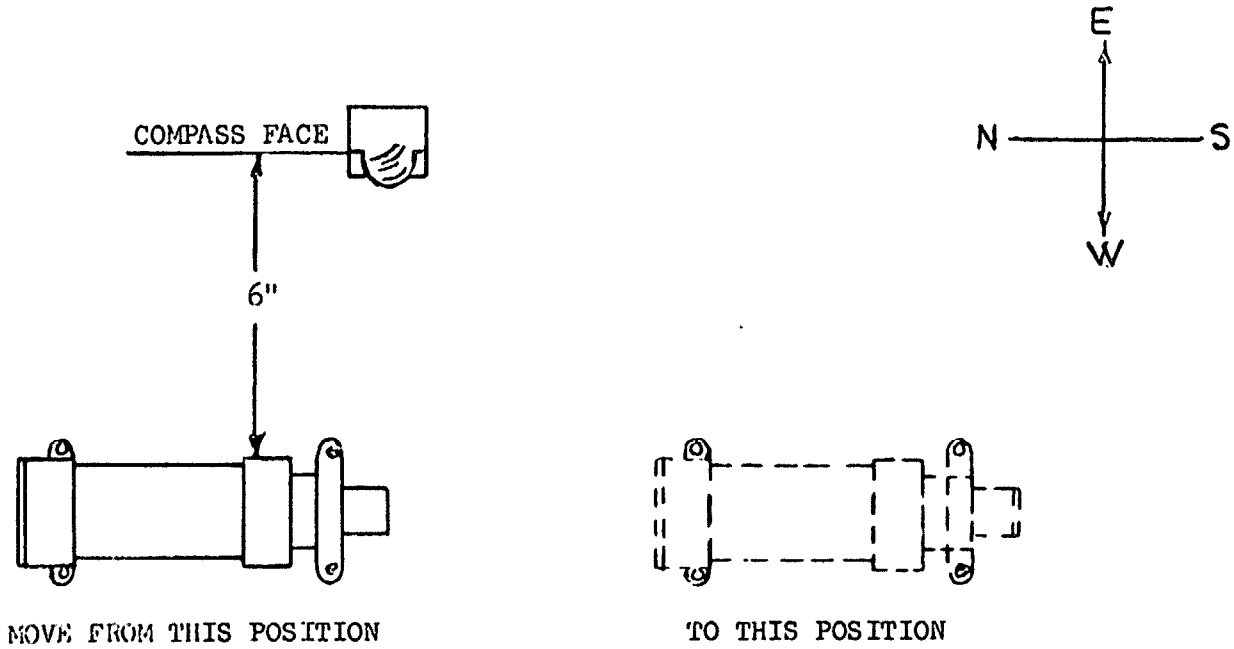
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Preparing activity:

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Project No. 1377-A252

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PLAN VIEW

NOTE: ACTUATOR, ROTARY, ASSEMBLY AND COMPASS SHALL LIE IN A HORIZONTAL PLANE

LAYOUT FOR DETERMINATION OF RESIDUAL MAGNETISM IN ACTUATOR, ROTARY, ASSEMBLY 1000-45

FIGURE 1

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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